

Small Scale Tuned Mass Damper

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Group 2

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Abstract

This experiment evaluated how well single and double pendulum setups worked as tuned mass dampers (TMDs) to protect tall buildings from earthquake damage. Using a PASCO wave driver, we simulated seismic shaking at 1.44 Hz and 1.63 Hz on a laser cut building model. We tested four different setups: an unchanged building (our control case), a single pendulum matched to the driving frequency ($\omega_{\text{driver}} = \omega_{\text{resonator}}$), a single unmatched pendulum ($\omega_{\text{driver}} \neq \omega_{\text{resonator}}$), and a double pendulum configuration that accounted for two frequencies at the same time. Two accelerometers tracked the movement at the base and top of the structure, from which the Root Mean Square (RMS) acceleration was calculated to measure the vibration energy of the system. The single matched pendulum minimized shaking through destructive interference, whereas the unmatched configuration amplified vibrations beyond the baseline. The double resonator setup successfully reduced the structural flex across both frequencies, but it caused a spike in energy transmission because the two pendulums started swinging with respect to each other. However, because this energy spike was minimal during the low frequency test, the double pendulum proved to be effective for low frequency stabilization but inefficient at higher frequencies.

Introduction

Earthquakes often transfer great amounts of kinetic energy to high-rise buildings through low-frequency seismic waves, causing dangerous structural damage and shear strain. To mitigate this damage, tuned mass dampers (TMDs) were made to absorb the energy in a controlled manner. In this experiment we tested the stabilization of a structure subjected to low frequency seismic waves while using simple pendulum resonators which acted as the TMDs. The theoretical angular frequency of these pendulums is defined by $\omega = \sqrt{\frac{g}{L}}$, where g is gravitational acceleration and L is the length of the pendulum. To quantify the energy trends of the structural vibrations and isolate amplitude magnitudes, we process the experimental acceleration data by calculating the Root Mean Square (RMS) for a discrete signal, mathematically defined as $x_{\text{RMS}} = \sqrt{\frac{1}{N} \sum x[n]^2}$ (for $\sum 1$ to N), where N is the number of terms and $x[n]$ is the n th term in the data.

The primary goal of this experiment is to determine how to minimize a building's structural flex during a seismic event without spiking its transmission amplitude. To achieve this, we used a PASCO wave driver to simulate high and low-frequency earthquakes on a structural building model. By measuring acceleration at the base and top of the structure with an ESP-32 microcontroller setup, we analyzed four distinct configurations: an unaltered base case, a single matched resonator, a single unmatched resonator, and a double resonator, to make a sum total of eight cases over both frequencies and 16 data sets over both ESP modules supplemented by six baseline calibration files.

We hypothesize that a single resonator tuned to precisely match the seismic driver frequency ($\omega_{driver} = \omega_{resonator}$) would stabilize the building, while an unmatched single resonator ($\omega_{driver} \neq \omega_{resonator}$) would cause even more destabilization than if we left it unaltered. Additionally, we expect that in a double-resonator system, there will generally be better stability than an unaltered building, regardless of whether the frequencies are matched or unmatched. However we also expect that the double resonant setup will be more prone to energy artifacts due to harmonic coupling between the resonators, but are unsure how it will affect our metrics. The following sections of this report will outline our experimental methods, data analysis, and overall significance of our findings.

Method

Initial Assumptions & Experimental Set up

Certain assumptions were established prior to starting our experiment. Namely, friction between the 3D-printed slides and the smooth table surface was assumed to be negligible, and the PASCO wave driver was assumed to operate as an ideal linear oscillator undergoing simple harmonic motion without any significant backdrive.

The setup consisted of several laser cut pieces for our building and two ESP-32 microcontrollers with accelerometer modules to record kinematic data. To power the ESP-32s, portable battery packs were attached directly using rubber bands. The resonators consisted of 100 and 200g masses attached via high tensile string to slotted pin-holes on the top of the building.

Procedure of Data Collection

The experimental apparatus was assembled starting with 3D-printed blocks placed directly on a smooth table surface. While originally intended as wheels, these blocks acted as low-friction sliders rather than rollers, supporting a thin, flat wooden board that served as the structural ground plane. The constructed laser cut building model was then placed completely unattached on top of this wooden surface. To simulate seismic waves, a PASCO wave driver was attached directly to the wooden board, acting as a linear shaker that vibrated the ground plane, which shook the building model resting on top of it as shown in figure 1.

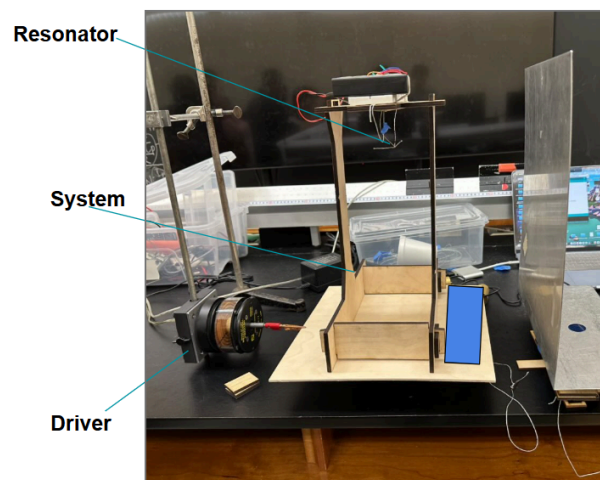


Figure 1: Experimental Setup

Before starting the data collection, the desired frequencies (1.44 Hz and 1.63Hz) were measured and calculated, and the high tensile string was cut to the corresponding lengths. Two ESP-32 microcontrollers equipped with accelerometer modules were connected through bluetooth to two separate computers to record data. One accelerometer was fixed to the base of the ground plane to capture the driver vibrations, while the other was mounted and secured using rubber bands to the roof of the building to track structural response. Because the data was being received on two different computers, global timestamps were turned on in the serial monitor so that the separate data sets from the top and bottom of the structure could be precisely synchronized by time during the data analysis.

Once the accelerometers were calibrated, the testing began with the base control case, where no pendulum resonator was attached to the building. This trial measured the building's baseline vibrations, representing the worst-case scenario for the structure. Following the control trial, two separate pendulum strings, cut to precise lengths for the target frequencies, remained suspended from the top of the tower, through the testing.

For the single resonator phase, a 200g mass was hooked onto the corresponding string. This phase included a matched case where the PASCO driver frequency was equal to the attached pendulum's natural frequency ($\omega_{driver} = \omega_{resonator}$), and an unmatched case where the driver frequency was out of sync ($\omega_{driver} \neq \omega_{resonator}$). The experiment then proceeded to the double resonator configuration where two 100g masses were hung from both strings simultaneously. Visually, this configuration seemed to be able to handle two frequencies at once, but the single matched case seemed to be more stable.

Procedure of Data Analysis

From the 16 data files received, a python algorithm first extracted and organized the data using a .txt parser function that extracted and reformatted the data. This included converting the text files into numpy arrays consisting of 5 columns: local time (ms), x_acc, y_acc, z_acc, global timestamp (converted to ms). As the various files had differing rowcounts, the overall data configuration required object based lists, with filename based access. After calibrating the accelerometers, using the 6 acceleration_calibration files, the data was constrained using a percentile diff function to remove the signal noise that arose from the edge artifacts of our building, and isolated using RMS to give a singular value for each of the 16 data_instances.

RMS, as an analytic function describing the scaled energy of a sinusoidal waveform, was used to formulate stability metrics. Consider y_1 to be the incoming signal of the ESP attached to the top, and y_2 be that of the ground ESP (driver). $RMS(y_1 - y_2)$ therefore indicates the energy of the relative motion of the top with respect to the bottom, and acts as a metric for building flex / structural deformation. High flex would mean the top was highly accelerated compared to the base, leading to high shear strain over time.

The other metric was $(RMS(y_1)/RMS(y_2))$, to indicate transmission as it showed the relative increase in energy from the driver wave to the resultant wave. However, due to the separate frequencies in oscillation despite having the same amplitude, the base cases of transmission metrics showed differing energies. To better contrast energy trends, the residuals of the transmission metric were taken. A high transmission ratio indicates a highly destabilized building as the amplitude of the driver wave is excessively magnified, and could result in 'tipping-over' if the center of gravity is critically displaced.

Results

The results of flexion and transmission showed a high correspondence to our first hypothesis regarding the stabilization of the matched single resonator case, but appeared to diverge with regards to our second hypothesis on the stabilization offered by the double resonator.

Data Calibration and Syncing

As evident from Figure 2, it is not possible to directly extract stability factors from what is given above. Therefore we first constrain the data by removing the percentile > 98% and < 2% error points, and then isolate the RMS as an analytical quantity from each of the 16 calibrated samples.

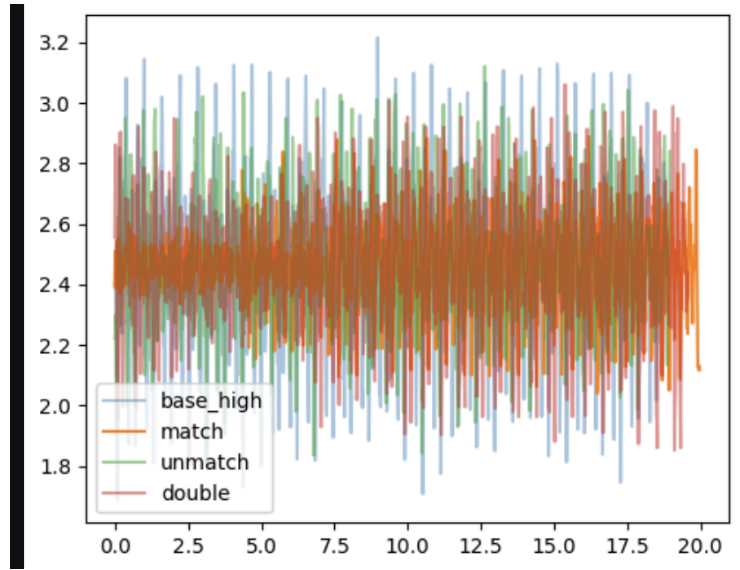


Figure 2: Calibrated and synced data sample (acceleration-time) for various configurations at high_driver frequency

Table 2: Raw RMS for 16 Data sets [m/s^2]

RMS(System) RMS(Driver) Raw values	
=====	
Base High	Sys RMS: 11.529 m/s^2 Drv RMS (Cal): 2.214 m/s^2
Base Low	Sys RMS: 9.347 m/s^2 Drv RMS (Cal): 2.217 m/s^2
Matched High	Sys RMS: 12.178 m/s^2 Drv RMS (Cal): 2.215 m/s^2
Matched Low	Sys RMS: 8.056 m/s^2 Drv RMS (Cal): 2.209 m/s^2
Unmatched High	Sys RMS: 10.940 m/s^2 Drv RMS (Cal): 2.211 m/s^2
Unmatched Low	Sys RMS: 10.444 m/s^2 Drv RMS (Cal): 2.227 m/s^2
Double High	Sys RMS: 14.543 m/s^2 Drv RMS (Cal): 2.141 m/s^2

Stability Metric 1 - Flex

As observed in figure 3,

1. There is a visible dip in the energy in the Matched cases compared to the unmatched case.
2. The dip in the matched case is much higher with the high driver. i.e. matched resonators stabilize higher frequencies better.
3. The double resonators also stabilize flexion better than base case, however the stabilization of the low-driver is much better.

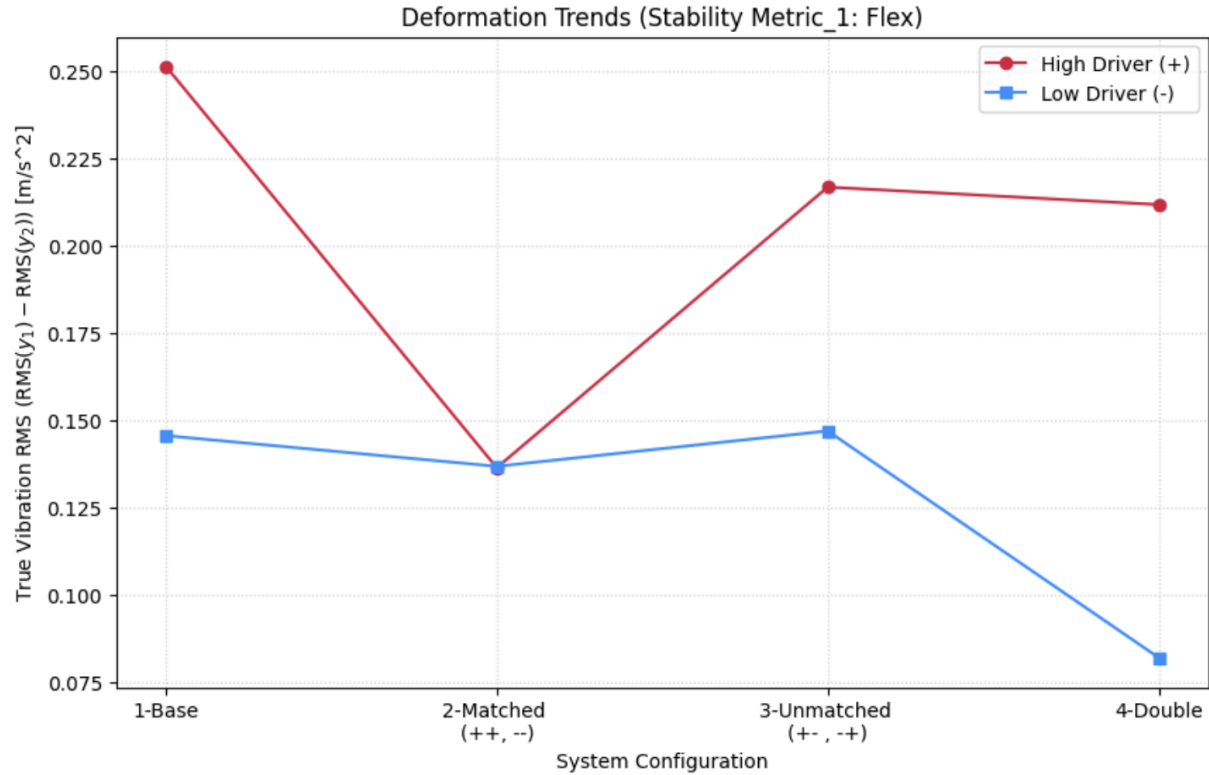


Figure 3: Flex: (RMS(y1) - y2) relative acceleration

The red and blue lines correspond to the high_driver frequency and low_driver frequency cases respectively, while the x axis describes the system configurations.

Table 3: Flexion [m/s²]

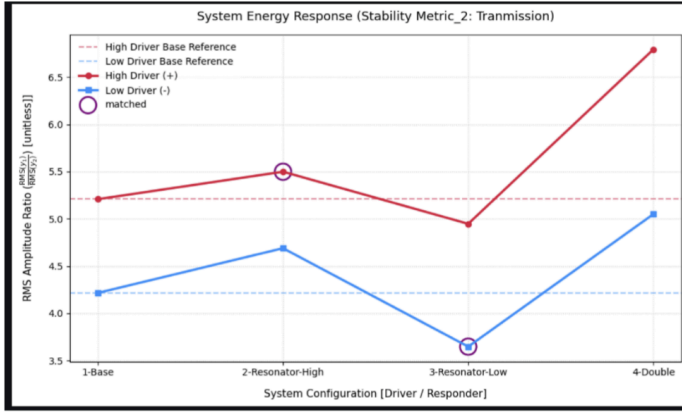
Relative Acceleration (Flexion of structure)		
base	Driver-High: 0.251 m/s ²	Driver-Low: 0.145 m/s ²
high	Driver-High: 0.136 m/s ²	Driver-Low: 0.147 m/s ²
low	Driver-High: 0.217 m/s ²	Driver-Low: 0.137 m/s ²
double	Driver-High: 0.212 m/s ²	Driver-Low: 0.082 m/s ²

Stability Metric 2 - Transmission

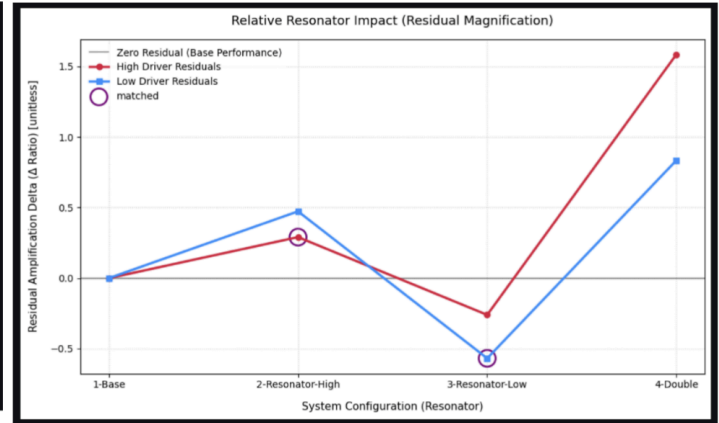
As is evident from Figure 4(b),

1. Compared to the overall energy trend, there are dips in the transmission
2. Both double resonant systems cause a large increase in energy transmission. This peak is relatively higher for the high-driver. (It is likely due to a harmonic mode created between the resonators.)

NOTE: The reason the base case RMS are different for high and low drivers, is because despite having the same amplitude, the acceleration wave form's amplitude is $A\omega^2$. Therefore the higher driver has more base energy/acceleration.



(a) Raw RMS Ratios ($\frac{RMS(y_1)}{RMS(y_2)}$)



(b) Residuals of RMS ratios

Figure 4: Transmission (Amplitude Magnification)

The purple circles correspond to the matched cases, while the red and blue line correspond to the high and low driver frequencies respectively.

Table 4: Transmission metrics [unitless ratios]

RMS magnification ratio ($RMS(y_1) \angle RMS(y_2)$) [transmission]			Relative Transmission (Residual Delta from Base case)		
1-Base	Driver-High: 5.208	Driver-Low: 4.216	1-Base	Driver-High: 0.000	Driver-Low: 0.000
2-Resonator-High	Driver-High: 5.499	Driver-Low: 4.689	2-Resonator-High	Driver-High: 0.291	Driver-Low: 0.473
3-Resonator-Low	Driver-High: 4.947	Driver-Low: 3.648	3-Resonator-Low	Driver-High: -0.260	Driver-Low: -0.568
4-Double	Driver-High: 6.793	Driver-Low: 5.050	4-Double	Driver-High: 1.585	Driver-Low: 0.834

Overall Results

For the single resonator systems,

the unmatched case is less stable than the base case while the matched case is more stable, in terms of both flex and transmission. The degree of added stability is higher for the higher driver frequency when considering both metrics. This provides two-fold confirmation of hypothesis A.

For the double resonator system,

the flex is stabilized for both driver frequencies, but the overall transmission is increased. However, for low-driver frequency, the flex stabilization is quite high, while the increase in transmission is relatively low. Therefore the double resonator setup best stabilizes low driver frequencies. But for higher frequencies, hypothesis B does not seem to hold merit due to the large spike in transmission.

Error Propagation

As RMS is an analytic function, the errors are dependent primarily on the sampling rates. As the ~sampling of our accelerometer was 67Hz (0.015s gap between readings) and our expected frequencies ranged between 0.5 and 3Hz, the data is likely highly accurate.

Random instrumental noise δx from the accelerometer modules propagates through the discrete RMS calculation as $\delta x_{RMS} \approx \frac{\delta x}{\sqrt{N}}$. Because each trial captured a massive dataset ($N \gg 1000$), statistical error propagation significantly suppresses the variance of the final calculated RMS values.

Discussion

Our results confirmed our first hypothesis but only conditionally satisfied the second. In the single-resonator setups, the RMS acceleration data shows that amplitude matching successfully minimized both flex and transmission energy through destructive interference. Furthermore, the unmatched single resonator amplified the leftover energy, destabilizing the structure as we predicted. The double-resonator system successfully reduced flex, especially under low-frequency drivers, but caused an unexpected spike in transmission amplitude. However, because the flex reduction was significant while the transmission increase was relatively small for the low-frequency driver, we determined that the double pendulum provided better overall stabilization under low-frequency. The waveform destabilization observed in the double resonator setup most likely occurred because of harmonic modes, with the two pendulums resonating with respect to each other.

Although our 16 data sets established clear analytical trends, there were some experimental uncertainties that impacted our findings. For example, the PASCO wave driver had low torque which caused jagged acceleration data at extreme points. Additionally, there were minor lateral forces causing the pendulum to swing off of our axis, affecting our accelerometers' readings. Finally, relying on global timestamp syncing may have caused marginal time drift between the top and bottom sensors.

If this experiment were repeated, these uncertainties could be minimized by using a linearly actuated high-torque shaker, confining the structural apparatus to a 2D plane via CNC alignment, and implementing an ESP-NOW protocol for precise microsecond byte-matching. To further explore our scientific question regarding the balance of flex and transmission, future experiments could investigate the use of active, sensor-driven tuned mass dampers. Dynamically adjusting dampening coefficients could potentially suppress the harmonic modes observed in the double-pendulum setup, maintaining a stable center of mass, without many downsides.

Conclusion

The primary objective of this experiment was to evaluate the stabilization of a structural building model under low-frequency seismic waves, when using tuned mass dampers. We hypothesized that a single resonator tuned to match the driving frequency ($\omega_{driver} = \omega_{resonator}$) would optimally stabilize the structure, whereas an unmatched resonator would further destabilize it. We also predicted that a double-resonator system would provide overall stability improvements over both frequencies. Our experimental results confirmed our first hypothesis twice – for both driver frequencies of the single-resonator systems, the matched case was much more stable than the base case in terms of both flex and transmission, while the unmatched case was less stable. As for our second hypothesis the double-resonator system produced mixed results. Despite the increased stability in flex for both driver frequencies, the structure's transmission was also greatly boosted. We ultimately determined that the double pendulum setup provides the best overall stabilization when subjected to a low-frequency driver, since it maintains high flex stability with only a low increase in transmission. For higher frequencies, however, the predicted stability improvements of the double-resonator system did not hold much merit. The significance and validity of these findings are firmly supported by our data analysis techniques. By ensuring the raw accelerometer data was calibrated across all 16 experimental trials, constrained, and then isolated using RMS, we can confidently conclude that our results are reliable indicators of overall stability trends. In summary, our results confirmed our first hypothesis for both frequencies, while our second hypothesis was only conditionally confirmed (for low frequencies).

References

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Report Timeline, Table (2)

Section	Range of dates drafted or done	Range of dates revised	Contributors to draft or activity	Contributors to revision
Title, Author, and Abstract	6/6-6/7	6/6-6/7	Jesus	Arnav, Raymond
Introduction and References	6/1-6/6	6/5-6/7	Raymond	Arnav
Methods	6/6-6/7	06/07	Jesus	Arnav, Raymond
Results	05/15 - 05/31	-	Arnav, Raymond, Jesus	Raymond
Data acquisition	05/15 - 05/28	05/26 - 06/02	Arnav, Raymond, Jesus	Jesus
Data analysis	05/26 - 05/30	05/29 - 06/04	Arnav	Raymond, Jesus
Conclusion	6/1-6/6	06/07	Raymond	Arnav, Jesus

Additional comments

Due to the nature of our data analysis, Error propagation could not be done to get discrete values. However an explanation as to why it is not directly achievable, as well as the relative significance of our data has been mentioned in the report.